

INVA SIGHT

Investment Data Pipeline Proposal

Data Engineering Bootcamp

Capstone Project

Team Members:

Maryam Alotaibi

Fayhaa Alharbi

Rawan Alaklabi

Hanoof Alassiri

Table Of Content

1. Introduction	3
2. Overview About WealthTech Industry	3
3. Problem Statement	3
4. Project Importance	3
5. Saudi Arabia WealthTech Market Analysis	3
6. Proposed Solution	4
7. Target Users	4
8. Data Sources	4
8.1 Internal Ledger Data Sample	5
8.2 Public Equity Market Feeds Sample	5
8.3 Foreign Exchange Reference Rates Sample	6
8.4 Precious Metals Market Rates Sample	6
9. Solution Design	6
10. Data Engineering Decisions	7
11. Downstream Use Case	8
12. Project Tools	8
13. Development Plan	9
14. Feasibility	9
15. Conclusion	9
16. References	10

1. Introduction

This report proposes the Investment Data Pipeline, a Data Engineering capstone project that automates the ingestion, standardization, and consolidation of investment data from four different sources into a single SAR-denominated dataset. It outlines the problem, data sources, pipeline design, and the team's delivery plan for the bootcamp timeframe.

2. Overview About WealthTech Industry

WealthTech, representing the intersection of wealth management and financial technology, refers to digital infrastructure and automated software solutions designed to enhance investment management, financial accessibility, and portfolio operations. Falling under the broader FinTech umbrella, modern WealthTech solutions focus on replacing legacy spreadsheet-driven manual workflows with automated data pipelines that process continuous market feeds and real-time asset pricing. A primary operational driver in the global WealthTech sector is the demand for unified portfolio management and reporting tools, with client portfolios increasingly spanning multiple asset classes, including commodities, which are regaining importance among investors as hedges and diversifiers, alongside traditional equities [2]. Modern WealthTech platforms aggregate multi-currency information feeds, including equities, foreign exchange, and spot commodities, into a single consolidated view to model the impact of exchange rate and share price fluctuations on client net worth.

3. Problem Statement

Investment data such as holdings, market prices, and FX rates are scattered across systems, currencies, and formats. Manually combining it to calculate daily portfolio value is slow, inconsistent, and error prone.

4. Project Importance

Manual and spreadsheet-reliant portfolio valuation cannot be scaled reliably when tracking diverse asset classes across global currencies. Industry research highlights backend data infrastructure, rather than front-facing tools, as the fundamental prerequisite for meaningful digital transformation in wealth management [1]. This project addresses this systemic operational gap by deploying an automated, repeatable data pipeline while enabling the team to gain practical experience in multi-source ingestion, strict data quality checks, and dimensional database modeling.

5. Saudi Arabia WealthTech Market Analysis

Saudi Arabia's asset management and FinTech sectors are experiencing rapid growth. This expansion is driven by locally managed assets under management surging to 1 trillion SAR by the end of 2024, alongside rapid FinTech growth, with active companies

reaching 261, surpassing initial targets and progressing toward the 2030 national goal of 525 firms [3]. This creates a clear need for reliable data engineering solutions that deliver a clean, standardized, and analytics-ready portfolio dataset.

6. Proposed Solution

An automated pipeline that ingests holdings, equity prices, and FX rates, converts everything to SAR, and outputs a clean, structured dataset ready for reporting and analytics.

7. Target Users

This project is designed for financial institutions, specifically targeting **wealth management teams** who need a consolidated and reliable view of their clients' portfolios. Individual investors may also derive secondary benefits from these insights.

8. Data Sources

The finalized data sources have provided us with what we need as a data engineer to formalize the required datasets for this project, here are our choices:

	Assets Class	Format	Access Method	Source Links
1	Internal Ledger Data	Synthetic CSV	Locally generated	Null
2	Public Equity Market Feeds	Live API	Daily stock/ETF prices via Alpha Vantage API	https://www.alphavantage.co/
3	Precious Metals Market Rates	Live API	Daily spot prices for commodities via metalprice API	https://metalpriceapi.com/
4	Foreign Exchange Reference Rates	Live API	Daily currency exchange rates via exchangerate API	https://www.exchangerate-api.com/docs/free

8.1 Internal Ledger Data Sample

This feed contains synthetic client transaction records capturing buy and sell activity, such as tickers, quantity, price, currency, and fees, to represent the firm's internal book of holdings.

transaction_id	client_id	client_name	portfolio_id	ticker	transaction_type	quantity	price	currency	fee_amount	transaction_date	total
T00001	C018	Khalid	P004	VOD	SELL	22	125.68	GBP	7.03	8/30/2026	-2757.93
T00002	C011	Khalid	P008	2222.SR	BUY	284	100.39	SAR	109.47	8/17/2026	28620.23
T00003	C006	Saad	P008	2222.SR	BUY	350	182.5	SAR	142.24	8/17/2026	64017.24
T00004	C010	Sara	P005	SAP	SELL	469	141.52	EUR	315.91	8/22/2026	-66056.97
T00005	C006	Omar	P002	IBM	BUY	362	185.87	USD	206.41	8/29/2026	67491.35
T00006	C012	Saad	P007	2222.SR	SELL	255	66.06	SAR	75.56	8/18/2026	-16769.74
T00007	C008	Huda	P008	XAU	SELL	484	436.5	USD	507.2	8/14/2026	-210758.8
T00008	C011	Aisha	P004	NESN.SW	SELL	54	392.25	CHF	61.74	8/9/2026	-21119.76
T00009	C007	Sara	P003	XAG	BUY	308	482.55	USD	636.55	8/3/2026	149261.95
T00010	C004	Omar	P007	XPT	BUY	171	266.98	USD	136.69	8/5/2026	45790.27
T00011	C001	Sara	P010	SAP	BUY	480	406.38	EUR	486.76	8/8/2026	195549.16
T00012	C019	Amal	P009	GOOGL	BUY	35	252.88	USD	9.51	8/23/2026	8860.31
T00013	C010	Faisal	P009	SAP	SELL	109	466.73	EUR	145.33	8/19/2026	-50728.24
T00014	C010	Maha	P003	IBM	SELL	318	179.53	USD	173.35	8/1/2026	-56917.19
T00015	C014	Rana	P008	NESN.SW	BUY	9	382.81	CHF	13.64	8/7/2026	3458.93
T00016	C006	Huda	P009	MSFT	SELL	228	348.7	USD	124.15	8/24/2026	-79379.45
T00017	C016	Reem	P001	VOD	SELL	215	233.15	GBP	66.75	8/3/2026	-50060.5
T00018	C014	Amal	P003	VOD	BUY	318	493.64	GBP	426.43	8/12/2026	157403.95
T00019	C011	Amal	P005	GOOGL	SELL	106	74.61	USD	12.79	8/12/2026	-7895.87
T00020	C010	Reem	P008	AAPL	SELL	145	109.1	USD	24.72	8/30/2026	-15794.78
T00021	C005	Aisha	P003	MSFT	BUY	98	304.81	USD	48.74	8/15/2026	29920.12
T00022	C015	Mohammed	P009	MSFT	SELL	371	80.2	USD	69.16	8/12/2026	-29685.04
T00023	C012	Huda	P001	AMZN	BUY	239	221.68	USD	220.65	8/28/2026	53202.17
T00024	C017	Amal	P002	XAU	SELL	411	388.71	USD	287.85	8/4/2026	-159471.96
T00025	C007	Nasser	P009	MSFT	SELL	266	371.46	USD	124.67	8/3/2026	-98683.69
T00026	C020	Amal	P010	HSBA.L	BUY	416	74.5	GBP	86.95	8/26/2026	31078.95
T00027	C015	Abdullah	P006	IBM	BUY	77	78.65	USD	19.38	8/27/2026	6075.43
T00028	C012	Ibrahim	P008	MSFT	SELL	254	178.18	USD	93.15	8/4/2026	-45164.57
T00029	C005	Saad	P001	AAPL	BUY	46	258.88	USD	25.68	8/14/2026	11934.16

8.2 Public Equity Market Feeds Sample

This feed contains historical daily close price data for public equities along with timestamps and stock tickers to track price movements over time.

date	open	high	low	close	volume	ticker
4/8/2026	258.45	259.7499	256.53	258.9	41032772	AAPL
4/9/2026	259	261.12	256.07	260.49	28121574	AAPL
4/10/2026	259.98	262.19	259.0231	260.48	31291473	AAPL
4/13/2026	259.73	260.18	256.66	259.2	36234698	AAPL
4/14/2026	259.245	261.93	257.19	258.83	48370710	AAPL
4/15/2026	258.16	266.56	257.81	266.43	49913510	AAPL
4/16/2026	266.8	267.16	261.27	263.4	43323112	AAPL
4/17/2026	266.96	272.3	266.72	270.23	61436228	AAPL
4/20/2026	270.33	274.275	270.29	273.05	36590169	AAPL
4/21/2026	271.5	272.8	265.4	266.17	50209755	AAPL
4/22/2026	267.82	273.74	266.87	273.17	43249204	AAPL
4/23/2026	275.045	275.77	271.65	273.43	33399639	AAPL
4/24/2026	272.755	273.06	269.65	271.06	38157110	AAPL
4/27/2026	266.09	268.36	265.07	267.61	41466762	AAPL
4/28/2026	272.335	273.23	268.66	270.71	40018940	AAPL
4/29/2026	267.55	271.04	267.04	270.17	30047869	AAPL
4/30/2026	270.5	276	268.14	271.35	91848230	AAPL
5/1/2026	278.855	287.22	278.37	280.14	79915442	AAPL
5/4/2026	279.655	280.63	274.8601	276.83	46668401	AAPL

8.3 Foreign Exchange Reference Rates Sample

This table details currency conversion reference rates relative to Saudi Riyal (SAR) for major currencies.

currency	date	rate_to_sar
USD	8/25/2026	3.75
EUR	8/25/2026	4.374584
GBP	8/25/2026	5.112571
CHF	8/25/2026	4.674069

8.4 Precious Metals Market Rates Sample

This market feed records closing prices for key spot commodities and precious metals denominated in foreign currency.

ticker	name	date	close_price	currency
XAU	Gold	2026-08-30	4455.55	USD
XAG	Silver	2026-08-30	66.37	USD
XPT	Platinum	2026-08-30	1827.03	USD
XPD	Palladium	2026-08-30	1422.47	USD

9. Solution Design

The pipeline follows a simple flow of input, processing, and output, transforming fragmented multi-currency data into a unified, analytics ready portfolio dataset.

Stage	Description
Input	- Multi-currency holdings (internal ledger) - international equity prices - Daily FX rates
Processing	- Cleaning and standardizing records - Convert all valuations into a single base currency (SAR) - Align trading calendars across markets (weekends/holidays) - Validate via automated data quality checks such as schema validation, deduplication, and reconciliation
Output	A unified, structured portfolio dataset modeled as a star schema, ready to support downstream analytics and reporting.

The diagram below illustrates the end-to-end user journey, from uploading raw holdings data to viewing and downloading the consolidated portfolio results.

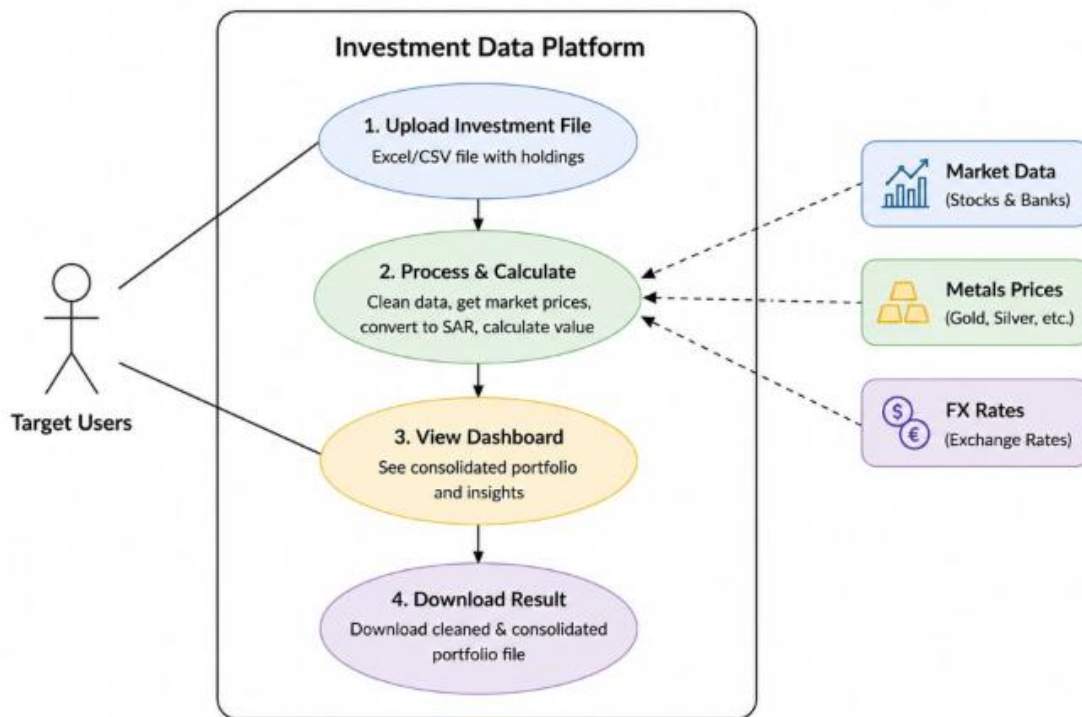


Figure 1 Shows a use case diagram

10. Data Engineering Decisions

To ensure the project delivers reliable, scalable, and analytics-ready data, we made the following key data engineering decisions:

- **ETL Batch Ingestion Architecture:** scheduled daily pipeline processing local holding exports alongside external market and FX price feeds
- **API Integration & Resilience:** automated extraction with fallback logic for missing feeds or rate limits
- **Dimensional Data Modeling:** a galaxy schema optimized for analytical query performance
 - dim_assets
 - fact_market_prices
 - fact_holdings
 - fact_daily_portfolio_summary
- **Data Quality Pipeline:** schema validation, non-null price filters, deduplication checks, and total valuation reconciliation rules

11. Downstream Use Case

The pipeline outputs a unified, structured portfolio dataset in SAR, modeled for BI dashboards, reporting systems, and investment analytics. It supports downstream use cases such as daily performance tracking and currency exposure reporting, consistent with core WealthTech "portfolio management" solutions (Jersey Finance, 2020) [1].

12. Project Tools

The following table presents the tools used throughout the pipeline's development and deployment.

Category	Tool(s)	Purpose
Data Platform	Snowflake	Stores the star schema (fact and dimension tables) and handles analytical queries
Data Extraction & Ingestion	Python (Pandas), pytest	Ingests multi-source market data, commodity feeds, and exchange rates; unit-tests extraction logic
Data Transformation & Testing	dbt	Transforms raw data into SAR-denominated star schema tables inside Snowflake
Workflow Orchestration	Apache Airflow	Schedules pipeline runs, manages task dependencies, and handles retries
Cloud Hosting & Infrastructure	AWS (EC2, S3)	Hosts raw data staging buckets, container registry, and managed infrastructure
Containerization	Docker	Packages extraction scripts and pipeline tasks into container images
Cluster Orchestration	Kubernetes	Deploys, manages, and scales containerized pipeline workloads on AWS
Version Control	Git	Tracks source code, dbt models, and pipeline configurations

13. Development Plan

The following table outlines the project timeline with throughout the bootcamp period:

No.	Phase	Key Deliverables
Week 1	Problem definition	Problem selection, source investigation, and access feasibility check
Week 2	Project Proposal	Finalized proposal document and slide deck.
Week 3	Pipeline & Data architecture design	Architecture diagram schema design and API extraction logic
Week 4	ETL Pipeline Development	Data extraction scripts and currency standardization to SAR
Week 5	ETL Pipeline Development	Star schema load and data quality rules
Week 6	Pipeline Testing & Evaluation	End-to-end pipeline validation, error handling, and BI integration
Week 7	Deployment & Final Presentation	Final dataset and demo

14. Feasibility

The pipeline uses synthetic local files and stable, free public APIs, avoiding scraping risk and licensing complexity while ensuring reliable execution. Asset reference metadata (ticker, currency, sector) is derived directly from the ledger and price feed responses within dim_assets, avoiding the need for a separate reference data source. The scope suits a four-person team, split across ingestion, transformation, modeling/quality, and orchestration.

15. Conclusion

This scalable, production-ready Data Engineering pipeline automates multi-asset financial data aggregation for the WealthTech domain. Replacing manual, error-prone reconciliation with an automated daily ETL workflow, the system normalizes global feeds, converts positions to SAR, and enforces strict data quality standards. The resulting Star Schema database provides wealth management teams and individual investors with daily portfolio visibility, risk monitoring, and audit-ready reporting.

16. References

- [1] Jersey Finance, "WealthTech put simply: An introduction to WealthTech and its impact on private wealth management," Tech. Rep., Sep. 2020. [Online]. Available: <https://www.jerseyfinance.com/wp-content/uploads/2020/09/JFL-Wealthtech-Put-Simply.pdf>
- [2] PwC Switzerland, "Wealth Management Insights 2026," Industry Rep., Mar. 2026. [Online]. Available: <https://www.pwc.ch/en/publications/2026/wealth-management-insights-2026.pdf>
- [3] Financial Sector Development Program, "2024 Annual Report," Council of Economic and Development Affairs (CEDA), Saudi Vision 2030. Available: https://www.vision2030.gov.sa/media/wpsn44ab/fsdp_annual-report-2024_-en.pdf